TEJAL BEDMUTHA

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EDUCATION

Columbia University, New York, NY — Sep 2025 – Dec 2026
M.S. in Computer Science
Coursework: Robot Learning, Deep Learning for Robot Manipulation, Machine Learning, Artifical Intelligence, Analysis of Algorithms, Databases.

PUBLICATIONS

- **Viewpoint-Invariant Manipulation via 3D Geometric Priors and Synthetic Data.**
  Tejal Bedmutha, Rishabh Jain, Judah Allen Goldfeder & Hod Lipson. | Under Review
- **A Survey on Sensor-based Planning and Control for Unmanned Underwater Vehicles.**
  Tejal Bedmutha & Shivam Vishwakarma. | Under Review
- **Instantaneous Planning, Control & Safety for Navigation in Unknown Underwater Spaces.**
  Veejay Karthik, Udit Ekansh, Tejal Bedmutha, Rohan Deshpande, Shivam Vishwakarma. | Under Review

WORK EXPERIENCE

Stealth Startup, New Jersey — Jun 2026 - Present.
Research Intern — NJ, USA.

- Working towards procedural synthetic-data pipeline for world-model pretraining, generating randomized 3D scenes, camera trajectories, object interactions, and physics-based rollouts for scalable vision-action learning.
- Designing RL post-training experiments for world-model agents; evaluated generation quality, interaction consistency, and goal completion across simulated environments.

Creative Machines Lab, Columbia University — Sep 2025 - Present.
Research Assistant | *Python, IsaacSim, Mujoco, LeRobot, Hugging Face.* Link — NYC, USA.

- Developed a viewpoint-invariant robot manipulation policy using 3D geometric priors and synthetic trajectory augmentation; evaluated against ACT under camera perturbations, improving robustness under unseen viewpoint shifts.
- Engineered a sim-to-real pipeline with multi-axis domain randomization: object poses, lighting, camera perturbations, and procedural dataset collection and generation pipeline, combined with Adversarial Motion Priors (AMP) to enforce naturalistic end-effector motion and reduce overfitting to demonstration artifacts, directly improving simulation accuracy.

Indian Institute of Technology, Bombay — May 2024 - Jun 2025
Research Assistant(Project Lead) | *ROS/ROS 2, Gazebo, Rviz, Nav2, Python, C++.* — Mumbai, India.

- Initiated and led development of a SONAR-based autonomy stack (mapping, motion planning, control) on a 200-ton AUV; independently integrated edge compute with sensor suites (SONAR, IMU, DVL) and drove end-to-end field deployment achieving 100m waypoint precision in GPS-denied environments.
- Led migration from ROS1 to ROS2 by redesigning node architecture and optimizing DDS communication, improving real-time system reliability under high-load conditions.
- Built a spatiotemporal mapping module atop OctoMaps, using KD-tree indexing and caching, and probabilistic occupancy persistence to classify static vs. dynamic obstacles 90% detection accuracy.

MIKO.AI — Jul 2023 - Apr 2024
Robotics Engineer. | *Raspberry Pi, NVIDIA Jetson, C++, Embedded C, Python.* — Mumbai, India.

- Built and debugged embedded firmware across sensors, motors, and hardware interfaces for a production robot that had to work reliably in uncontrolled environments
- Tracked down real-world failures across sensing, planning and control loops, not reproducible in simulation.
- Designed a lightweight LiDAR localization system that runs on constrained hardware, after standard pipelines failed latency constraints.
- Led bring-up of a new robotics platform from scratch (Miko Nano), integrating mechanical, electrical, and software systems

MITACS - GRI — May 2022 - Aug 2022
Research Intern. | *MATLAB, Gazebo, Simulink.* Link — Montreal, Canada.

- Designed and implemented a MATLAB simulation framework to develop and evaluate an efficient UAV clustering algorithm; optimizing multi-agent target assignment across configurable parameters, reducing clustering time by 20 seconds over conventional evolutionary methods.
- Validated clustering algorithm through Gazebo/ROS simulation and real hardware deployment ensuring an end-to-end pipeline.

TECHNICAL SKILLS

Languages	: Python, C++, C, Embedded-C, SQL, MATLAB.
Databases	: PostgreSQL, MongoDB, pgvector, FAISS.
AI / LLM	: PyTorch, TensorFlow, LangGraph, LlamaIndex, RAGAS, DeepEval, LoRA, PEFT, HuggingFace, OpenAI, Anthro
Libraries	: OpenCV, Scikit-Learn, NumPy, pandas, spaCy, sentence-transformers, LeRobot.
Simulation	: ROS/ROS2, MuJoCo, Gazebo, RViz, IsaacSim, CoppeliaSim, Simulink, Blender.
Edge / Infra	: NVIDIA Jetson, Raspberry Pi, embedded C, Docker, Linux, AWS, GCP, CI/CD, MLflow.
Developer Tools	: Git, Bash, Shell Scripting, pytest, Weights & Biases, Jupyter.